

Functional Analysis and Physics

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July 27, 2026

“It is at this point that normal language gives up, and goes and has a drink.”

- Terry Pratchett, *The Color of Magic*

“We are wandering at random, we are exerting ourselves only to return to the same spot, 'tis labor lost”

-Aristophanes, *The Birds*

Introduction/layout

- Most of this presentation is inspired by the discussion in chapter 1 of *Functional Integration*, by Cecilé Dewitt-Morette and Pierre Cartier
- We will discuss briefly the heat equation and the Schrödinger equation, then digress into functional analysis
- We will then discuss some applications of functional analysis in black hole thermodynamics

The heat equation

- The three dimensional analogue to the heat (diffusion) equation that we've seen is:

$$\frac{\partial \Psi}{\partial t} = D \nabla^2 \Psi.$$

- And the Green's function we've seen for this differential operator is:

$$G(\mathbf{r}, t) = \frac{1}{(4\pi Dt)^{3/2}} e^{-\mathbf{r}^2/4Dt}.$$

The Schrödinger equation

- The free particle Schrödinger equation is

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi.$$

- And the Green's function for this differential operator is:

$$G(\mathbf{r}, t) = \left(\frac{m}{2\pi\hbar it} \right)^{\frac{3}{2}} e^{\frac{im|\mathbf{r}|^2}{2\hbar t}}$$

- Note that by performing a “wick rotation” $t \rightarrow it$ and adjusting D , we can make the heat equation and Schrödinger equation “the same”.

Two Planck Constants

- Plank introduced two constants into physics in the late 1800's, both related to the energy E ,

$$E = \hbar\omega \qquad E = k_B T$$

- Where $\hbar$ is the reduced Planck constant and k_B is called the Boltzmann constant, so named by Planck. What is the relationship between these two constants?

Spaces of pointed paths with fixed initial and final points

- A space of pointed paths $\mathcal{P}_{a,b}$ with fixed initial point $a \equiv (x_a, t_a)$ and endpoint $b \equiv (x_b, t_b)$ is a complete normed vector (function) space. Elements of this space are called *paths*.

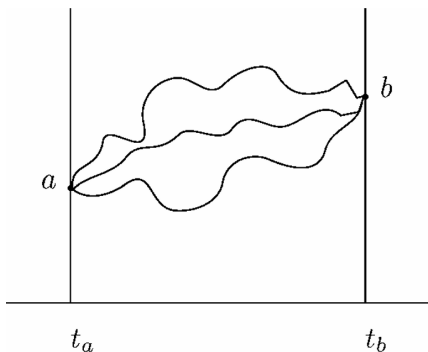


Figure: Space of paths with 3 elements

Functional Analysis

- A *functional* is a map $F : \mathcal{P}_{a,b} \rightarrow \mathbb{F}$, where $\mathbb{F}$ is a field, e.g. $\mathbb{R}$.
- The functional derivative (of functional F w.r.t. function f) is defined analogously to the ordinary derivative.

$$\frac{\delta F}{\delta f} \equiv \lim_{\epsilon \rightarrow 0} \frac{F[f(x') + \epsilon \delta(x - x')] - F[f(x')]}{\epsilon}$$

- The functional integral is trickier. It can be shown that it is a type of Lebesgue integral if the integrand is real, the complex case is an open question.

Back to physics: Hamilton's principle

- A dynamical system is completely defined by an *action functional* S . Locality implies that the action can be written as an integral over spacetime

$$S = \int \mathcal{L} d^4x + \text{boundary terms} \quad (1)$$

where $\mathcal{L} = \mathcal{L}(x, \dot{x}, t)$ is a lagrangian (density).

- A *dynamical history* is a stationary point of the action functional, i.e. a set of curves that satisfy *Hamilton's principle*

$$\delta S = 0 \quad (2)$$

- These paths form a subspace called the *dynamical subspace* or *dynamical shell*. The relativistic dynamical shell is sometimes called the mass shell since $\delta S = 0 \implies p^2 = -m^2$.

Quantum partition function and time propagator

- The quantum partition function is given by

$$Z(\beta) = \text{Tr} \left(e^{-\beta H} \right) = \sum_{\psi} \langle \psi(0) | e^{-\beta H} | \psi(0) \rangle$$

- This looks similar to the time evolution operator

$$U_t = e^{-itH/\hbar},$$

such that

$$|\psi(t)\rangle = U_t |\psi(0)\rangle = e^{-itH/\hbar} |\psi(0)\rangle$$

- How are these related? We'll come back to that.

Weiner's path integral

- We consider now the general (inhomogenous) form of the heat equation/Schrödinger equation, respectively corresponding physically to the existence of a heat source/potential energy. The general heat equation can be written

$$\frac{\partial \Psi}{\partial t} = D \nabla^2 \Psi - W(x) \Psi$$

- with solution given by *Kac's formula*

$$\psi(x_b, t_b) = \mathbb{E}_{a,b}[\exp[\int (-W(q(t))dt)]]$$

where $\mathbb{E}_{ab}$ is the integral over all sample paths of brownian motion with tied-up ends.

Feynman's path integral

- The general Schrödinger equation can be written

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi - V(\mathbf{r})\Psi.$$

- The Feynman path integral solution to the Schrödinger equation is

$$\Psi(x_b, t_b) = \int \mathcal{D}q \cdot \phi(q(t_a)) e^{\frac{i}{\hbar} S(q)},$$

where $\phi(q(t_a))$ is the wave function at a point $q(t_a)$ ($\psi(x, t_a) = \phi(x)$) and the integral is taken over the space of paths (not necessarily on shell!). $S(q)$ is the action,

$$S(q) = \int_{t_a}^{t_b} dt L(\dot{q}, q).$$

Equivalence of Feynman path integral with Wiener

- For both situations, we can write down a solution in terms of the kernel. For the detailed calculation, see [1]
- For the Wiener process, we have

$$K_{\beta}(x_b; x_a) = \int \mathcal{D}q \cdot \exp \left(-\frac{1}{\hbar} \int_0^{\beta \hbar} d\tau H(q(\tau), \dot{q}(\tau)) \right)$$

For Feynman's integral,

$$L_u(x_b; x_a) = \int \mathcal{D}q \cdot \exp \left(\frac{i}{\hbar} \int_0^u dt L(\dot{q}(t), q(t)) \right)$$

The connection

- Expressing the inner term of these previous equations, gets us

$$-\frac{1}{\hbar} \int_0^{\beta\hbar} d\tau H(q(\tau), \dot{q}(\tau)) = \int_0^{\beta\hbar} \left[-\frac{m}{2\hbar} \frac{|dq|^2}{d\tau} - \frac{1}{\hbar} V(q) d\tau \right],$$
$$\frac{i}{\hbar} \int_0^u dt L(\dot{q}(t), q(t)) = \int_0^u \left[\frac{im}{2\hbar} \frac{|dq|^2}{dt} - \frac{i}{\hbar} V(q) dt \right].$$

- These match if $\tau = it$ and $\beta\hbar = iu$, where $u \equiv -i\hbar\beta = -\frac{i\hbar}{k_B T}$ is the *imaginary Boltzmann time*

Imaginary Boltzmann time

- This is the relationship between the diffusion equation and the Schrödinger equation, the quantum partition function and the time evolution propagator, the relation

inverse temperature $\sim$ imaginary time.

- The i in the integral means “Feynman’s theory is not easily made rigorous. Indeed, one needs an integration theory more general than Lebesgue theory”¹

¹From the same book, *Functional Integration: Action and Symmetries*

$$c \neq \infty$$

- The aforementioned relation

$$Z = \text{Tr}[U(u)] = \text{Tr}[\exp[-\beta \hat{H}]]$$

imposes on a system in equilibrium a periodicity β in imaginary time ($\hbar = 1$ hereafter).

- The transformation $t \rightarrow it$ is called *Euclideanization* of the spacetime metric, because it corresponds with $\eta_{\mu\nu} \rightarrow \delta_{\mu\nu}$
- The Lorentz transformations become rotations of a 4D sphere (i.e. elements of $\text{SO}(4)$).

Schwarzschild Black Holes

- This section is borrowed from [4]. Birkhoff's theorem guarantees a unique spherically symmetric solution to Einstein's equation $G = T$, the *Schwarzschild solution*,

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} + r^2 d\Omega^2$$

where $r_s = 2GM$ is the *Schwarzschild radius*, $c = 1$, and $d\Omega^2 = \sin^2(\theta)d\phi^2 + d\theta^2$ is the angular part of the metric.

- To study the near horizon geometry, we introduce the coordinates

$$r = r_s(1 + \rho^2) \tag{3}$$

- The Euclideanized ($dt'^2 = -dt^2$) metric becomes

$$ds^2 \approx 4r_s^2(d\rho^2 + \rho^2(\frac{dt'}{2r_s})^2 + \frac{1}{4}d\Omega^2) \tag{4}$$

when $\rho \ll 1$.

Hawking Temperature

- Locally, the spacetime metric must be flat (an assumption of the manifold model for spacetime). In order for the metric

$$ds^2 \approx 4r_s^2(d\rho^2 + \rho^2(\frac{dt'}{2r_s})^2 + \frac{1}{4}d\Omega^2) \quad (5)$$

to be flat, the periodicity in dt' , already identified with β , must be $4\pi r_s$. That is,

$$\beta = 4\pi r_s \implies T = (8\pi MG)^{-1}. \quad (6)$$

- This is the so-called *Hawking Temperature*.

Bekenstein-Hawking entropy

- The Hamiltonian for the Schwarzschild black hole is just its mass. This suggests that the first law of thermodynamics takes the form

$$dM = TdS \implies S = 4\pi M^2 G \quad (7)$$

- Note that the area of the black hole is $A = 4\pi r_s^2$. Therefore,

$$S = \frac{A}{4G} \quad (8)$$

the so-called *Bekenstein Hawking entropy*.

References and further reading

- ① Cartier, P. & DeWitt-Morette, C. (2006). *Functional integration: Action and symmetries*. Cambridge University Press.
- ② Feynman, R.P. & Hibbs, A.R. (1965). *Quantum mechanics and path integrals*. Dover.
- ③ Sakurai, J.J. & Napolitano, J. (2020). *Modern quantum mechanics*. Cambridge University Press.
- ④ Becker, K., Becker, M. & Schwartz, J. (2007). *String Theory and M-Theory*. Cambridge University Press.